

COMP6030 – Sprint 5 meeting minutes

Date: Friday 13th February 2026

Sprint dates: Feb 9 – Feb 13 2026

Attendees: Jaden, Sattyaj, Sinead, Saf. All team members present

Sprint goals

Implement core pathfinding algorithm, calculate travel times, station selection UI

Tasks planned this sprint

- Implement pathfinding algorithm
- Unit test pathfinding with sample queries
- Implement station search
- Toggle station state
- Write Unit tests for scenario logic
- Implement station selection UI

Completed tasks and notes

- Dijkstra's algorithm implemented over the network graph. Edge weights are travel times in minutes. Research into how to use travel time ongoing
- Tests pass on small networks. Verified it finds the correct shortest path between a few well known routes (e.g. King's Cross to Waterloo).
- Scenario logic tests cover closing a station, attempting a route through it, verifying the system finds an alternative or returns no route found.
- Station state toggle implemented so stations can be opened or closed interactively as part of scenario setup.

Blockers / Issues

- Algorithm implementation as the biggest task this sprint took up most of the sprint but the algorithm is solid.
- Finding a way to find out or calculate travel time is pushed for next sprint but the algorithm works fine on distance for now but the intention is to move to travel times when data is obtained.
- Discussed whether to use A* instead of Dijkstra. We decided Dijkstra is fine for this network size. Can revisit in the future when looking at optimisation

Actions / Next steps

- Integrate routing into the backend API next sprint
- Handle edge cases (no route found, same start/end) in Sprint 6
- Add the option to close entire lines
- Research travel times and implement into algorithm
- Start showing the route on the map

